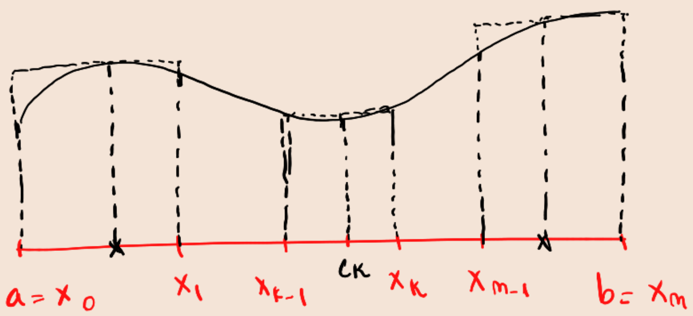
① Compute the following limits using Riemann sum.

Riemann sum:

Let $a = x_0 < x_1 < x_2 < \dots < x_k < \dots < x_m = b$.



– Goal: approximate the graph using integrals → area of □

$$\sum_{k=1}^{m} \underbrace{f(c_k)}_{\text{height of } \square} \underbrace{(x_k - x_{k-1})}_{\text{base of } \square \text{ (partition)}} \to \int_a^b f(x)\,dx, \quad c_k \in [x_{k-1}, x_k]$$

Ex: $[0,1]$

$x_k = \frac{k}{m}$, $0 < \frac{1}{m} < \frac{2}{m} < \dots < \frac{m-1}{m} < 1$

$c_k = \frac{k}{m}$

$$\frac{1}{m} \cdot \sum_{k=1}^{m} f\left(\frac{k}{m}\right) \to \int_0^1 f(x)\,dx$$

the simplest form of how a Riemann sum would look like.

a) $\lim\limits_{m} \sum\limits_{k=1}^{m} \frac{1}{m+k}$

$= \lim\limits_{m} \frac{1}{m} \sum\limits_{k=1}^{m} \frac{1}{1+\frac{k}{m}}$ → $f\left(\frac{k}{m}\right)$, $f(x) = \frac{1}{1+x}$

$= \int_0^1 f(x)\,dx$

$= \int_0^1 \frac{1}{1+x}\,dx = \int_0^1 \frac{(1+x)'}{1+x}\,dx = \ln|1+x|\Big|_0^1 = \ln 2 - \ln 1 = \ln 2.$

$\Rightarrow \frac{1}{m+1} + \frac{1}{m+2} + \dots + \frac{1}{2m} \to \ln 2$

(this sum converges to $\ln 2$)

b) $\lim_{n} \sum_{k=1}^{n} \frac{k^p}{n^{p+1}}, \ p \in \mathbb{N}.$

$$= \lim_{n} \frac{1}{n} \sum_{k=1}^{n} \left(\frac{k}{n}\right)^p = \int_0^1 x^p \, dx = \frac{x^{p+1}}{p+1}\Big|_0^1 = \frac{1}{p+1}$$

$f(x) = x^p$

c) $\lim_{n} \frac{\sqrt[n]{n!}}{n} = \lim_{n} \sqrt[n]{\frac{n!}{n^n}} = \lim_{n} \sqrt[n]{\frac{1}{n} \cdot \frac{2}{n} \cdot \ldots \cdot \frac{n}{n}}$

– I have a product, but I need a sum –

log (prod.) = sum (log)

$\log(a \cdot b) = \log a + \log b$

Take $x_n = \sqrt[n]{\frac{1}{n} \cdot \frac{2}{n} \cdot \ldots \cdot \frac{n}{n}}$

$$\ln x_n = \ln \left(\frac{1}{n} \cdot \frac{2}{n} \cdot \ldots \cdot \frac{n}{n}\right)^{\frac{1}{n}} = \frac{1}{n} \ln \left(\frac{1}{n} \cdot \ldots \cdot \frac{n}{n}\right) = \frac{1}{n} \sum_{k=1}^{n} \ln \frac{k}{n}$$

$f(x) = \ln x$

$$= \int_0^1 \ln x \, dx$$

– But $\int_0^1 \ln x \, dx$ is an improper integral because we cannot integrate directly in $0 \to \ln 0 = -\infty$ –

$$\int_0^1 \ln x \, dx = \int_0^1 x' \cdot \ln x \, dx = x \ln x \Big|_0^1 - \int_0^1 \cancel{x} \cdot \frac{1}{\cancel{x}} dx =$$

$$= 1 \ln 1 - \lim_{x \searrow 0} x \ln x - x \Big|_0^1$$

$$= 0 - 0 - 1$$

$$= -1$$

$$\lim_{x \searrow 0} x \ln x = \lim_{x \searrow 0} \frac{\ln x}{\frac{1}{x}} \overset{\frac{-\infty}{\infty}}{\underset{L'H}{=}} \lim_{x \searrow 0} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x \searrow 0} (-x) = 0.$$

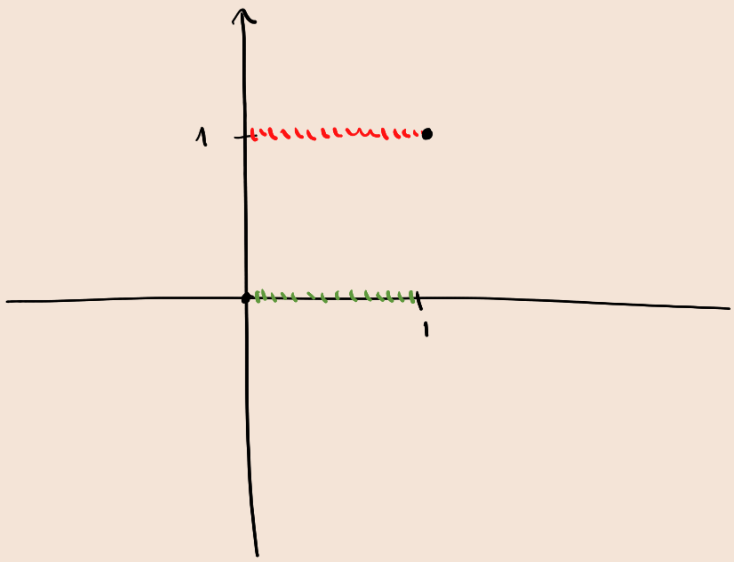
$\Rightarrow \ln x_n = -1 \Rightarrow x_n = \frac{1}{e}$

$\Rightarrow \lim_{n} \frac{\sqrt[n]{n!}}{n} = \frac{1}{e}.$

d) Study the integrability of $f: [0,1] \to \mathbb{R}$

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

– This function is discont. at any point –



Let $x_k = \frac{k}{n}$, $\quad \underline{x_0 = 0} < x_1 < \dots < x_k < \dots < x_{n-1} < \underline{x_n = 1}.$

(this is a partition)

$$\underbrace{(1+1+\dots+1)}_{n} \cdot \frac{1}{n} = n \cdot \frac{1}{n} = 1.$$

For $c_k \in [x_{k-1}, x_k]$ and $c_k \in \underline{\mathbb{Q}}$: $\displaystyle\sum_{k=1}^{n} \underbrace{f(c_k)}_{1} \underbrace{(x_k - x_{k-1})}_{\frac{1}{n}} = 1.$

For $c_k \in [x_{k-1}, x_k]$ and $c_k \in \underline{\mathbb{R} \setminus \mathbb{Q}}$: $\displaystyle\sum_{k=1}^{n} \underbrace{f(c_k)}_{0} \underbrace{(x_k - x_{k-1})}_{\frac{1}{n}} = 0$

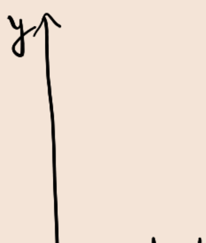
(for the same partition we took $\neq$ points)

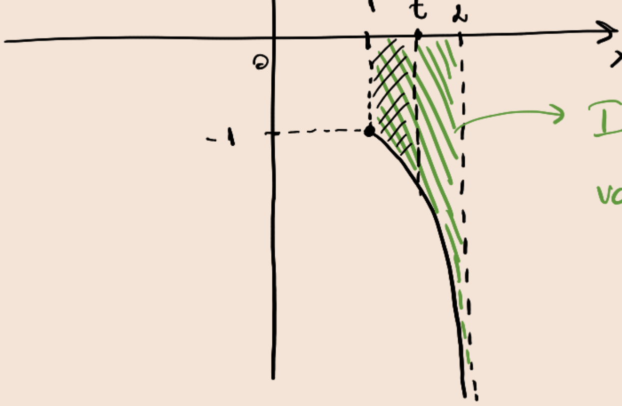
$\Rightarrow$

$\Rightarrow \nexists \displaystyle\int_0^1 f(x)\,dx \Rightarrow$ I can't integrate the function $\Rightarrow$ non-integrable.

② a) $\displaystyle\int_1^2 \frac{1}{x(x-2)}\,dx$

– Check the graph. If I don't have a vertical asymptote $\Rightarrow$ usual function $\Rightarrow$ integrable. If yes $\Rightarrow$ compute area –





I need to compute the value of this ∞ region

$$\lim_{t \nearrow 2} \int_1^t \frac{1}{x(x-2)}\, dx$$

$$\int_1^t \frac{1}{x(x-2)}\, dx = -\frac{1}{2} \cdot \int_1^t \frac{x-2-x}{x(x-2)}\, dx = -\frac{1}{2}\int_1^t \left(\frac{1}{x} - \frac{1}{x-2}\right) dx =$$

$$= -\frac{1}{2}\int_1^t \frac{1}{x}\, dx + \frac{1}{2}\int_1^t \frac{1}{x-2}\, dx = -\frac{1}{2} \ln|x| \Big|_1^t + \frac{1}{2} \ln|x-2| \Big|_1^t =$$

$$= -\frac{1}{2} \ln t + \frac{1}{2} \ln(2-x) \Big|_1^t = -\frac{1}{2} \ln t + \frac{1}{2} \ln(2-t)$$

$$\lim_{t \nearrow 2} \int_1^t \frac{1}{x(x-2)}\, dx = \lim_{t \nearrow 2} \left(-\frac{1}{2} \ln t + \frac{1}{2} \ln(2-t)\right) = -\frac{1}{2} \ln 2 - \infty = -\infty.$$

=> Conclusion: that ∞ region has ∞ area => divergent.

b) $$\int_0^{\infty} x \cdot e^{-x^2} dx = \lim_{t \to \infty} \int_0^t x \cdot e^{-x^2} dx = \lim_{t \to \infty} \rightarrow \text{Method 1}$$

the problem is in ∞

$$(x \cdot e^{-x^2})' = x' \cdot e^{-x^2} + x \cdot (e^{-x^2})' = e^{-x^2} + x \cdot e^{-x^2} \cdot (-2x) \ ?$$

$$(x \cdot e^{-x^2})' = -2x \cdot e^{-x^2}.$$

$$\int_0^{\infty} x \cdot e^{-x^2} \rightarrow \text{Method 2} = -\frac{1}{2}\int_0^{\infty} (e^{-x^2})'\, dx = -\frac{1}{2} e^{-x^2} \Big|_0^{\infty} = -\frac{1}{2} e^{-\infty} + \frac{1}{2} = \frac{1}{2}$$

c) $$\int_0^1 \frac{\ln x}{\sqrt{x}}\, dx = \lim_{t \searrow 0} \int_t^1 \frac{\ln x}{\sqrt{x}}\, dx \overset{(*)}{=} -4$$

$$2\cdot\int_t^1 \frac{\ln x}{2\sqrt{x}}dx = 2\cdot\int_t^1 (\sqrt{x})'\cdot \ln x\,dx = 2\left(\sqrt{x}\ln x\Big|_t^1 - \int_t^1 \cancel{\sqrt{x}}\cdot\frac{1}{\cancel{x}}\,dx\right) =$$

$$= 2\left(1\ln 1 - \sqrt{t}\ln t - 2\sqrt{x}\Big|_t^1\right)$$

$$= 2\left(1\ln 1 - \sqrt{t}\ln t - 2(1-\sqrt{t})\right)$$

$$= 2\left(0 - \sqrt{t}\ln t - 2 + 2\sqrt{t}\right)$$

$$= \underline{-2\sqrt{t}\ln t} - 4 + 4\sqrt{t}$$

$$\xrightarrow{t\searrow 0} 0 - 4 + 0 = -4.\ (*)$$

$$\int \frac{1}{\sqrt{x}} = \int x^{-\frac{1}{2}} = \frac{x^{1-\frac{1}{2}}}{1-\frac{1}{2}} = \frac{x^{\frac{1}{2}}}{\frac{1}{2}} = \frac{\sqrt{x}}{\frac{1}{2}} = 2\sqrt{x}$$

$$\lim_{t\searrow 0}\sqrt{t}\ln t = \lim_{t\searrow 0}\frac{\ln t}{\frac{1}{\sqrt{t}}} \overset{L'H}{=} \lim_{t\searrow 0}\frac{\frac{1}{t}}{-\frac{1}{2t\sqrt{t}}} = \lim_{t\searrow 0} -2\sqrt{t} = 0.$$

d) $$\int_0^\infty e^{-x}\cdot \sin x\,dx = \int_0^\infty (-e^{-x})'\cdot \sin x\,dx = I.$$

$$= \underline{-e^{-x}\cdot\sin x}\Big|_0^\infty + \int_0^\infty e^{-x}\cdot\cos x\,dx$$

$$= -\lim_{x\to\infty} \underbrace{e^{-x}}_{\searrow 0}\cdot\underbrace{\sin x}_{\text{bounded}} + 0 + \int_0^\infty e^{-x}\cdot\cos x\,dx$$

$$= 0 + 0 + \int_0^\infty e^{-x}\cdot\cos x\,dx$$

$$= \int_0^\infty (-e^{-x})'\cdot\cos x\,dx$$

$$= \underline{-e^{-x}\cdot\cos x}\Big|_0^\infty + \int_0^\infty -e^{-x}\cdot\sin x\,dx$$

$$= -\lim_{x\to\infty}\underbrace{e^{-x}}_{\searrow 0}\cdot\underbrace{\cos x}_{\text{bounded}} + 1 - \underbrace{\int_0^\infty e^{-x}\cdot\sin x\,dx}_{I}$$

$$\Rightarrow I = 0 + 1 - I$$

$$\Rightarrow 2I = 1 \Rightarrow I = \frac{1}{2}$$

(4) Study the convergence for:

a) $$\int_1^\infty \frac{1}{x\sqrt{1+x^2}}dx$$ $\rightarrow$ behaves like x^2.

convergent if $p>1$

Fact: $\int_1^\infty \frac{1}{x^p}\,dx$ → converg... if p
→ divergent if $p \le 1$

$$\lim_{x\to\infty} \frac{\frac{1}{x\sqrt{1+x^2}}}{\frac{1}{x^2}} = \lim_{x\to\infty} \frac{x^2}{x\sqrt{1+x^2}} = \lim_{x\to\infty} \frac{x}{\sqrt{1+x^2}} = \lim_{x\to\infty} \sqrt{\frac{x^2}{1+x^2}} = 1.$$

$\Rightarrow \int_1^\infty \frac{1}{x\sqrt{1+x^2}}\,dx$ „like" $\int_1^\infty \frac{1}{x^2}\,dx$ convergent.

b) $\int_0^{\frac{\pi}{2}} \frac{1}{\cos x}\,dx$

↳ the problem is in $\frac{\pi}{2}$

Fact: $\int_a^b \frac{1}{(b-x)^p}\,dx$ → conv. if $p<1$
→ div. if $p \ge 1$.

Compare $\frac{1}{\cos x}$ with $\frac{1}{\left(\frac{\pi}{2}-x\right)^p}$

$$\boxed{\cos x = \sin\left(\frac{\pi}{2}-x\right)}$$

$$\frac{1}{\cos x} = \frac{1}{\sin\left(\frac{\pi}{2}-x\right)}$$

$$\frac{\cos x}{\frac{\pi}{2}-x} = \frac{\sin\left(\frac{\pi}{2}-x\right)}{\frac{\pi}{2}-x} \to \lim_{x\nearrow\frac{\pi}{2}} \to 1.$$

$$\Rightarrow \frac{\frac{1}{\cos x}}{\frac{1}{\frac{\pi}{2}-x}} \to 1 \Rightarrow p=1.$$

$\int_0^{\frac{\pi}{2}} \frac{1}{\cos x}\,dx$ „like" $\int_0^{\frac{\pi}{2}} \frac{1}{\frac{\pi}{2}-x}\,dx$ divergent

$$\downarrow$$

$$\int_0^{\frac{\pi}{2}} \frac{2}{\pi-2x}\,dx = \int_0^{\frac{\pi}{2}} \frac{1}{2x-\pi}\,dx = \frac{1}{2}\int_0^{\frac{\pi}{2}} \frac{(2x-\pi)'}{2x-\pi}\,dx$$

$$= \frac{1}{2}\ln|2x-\pi|\Big|_0^{\frac{\pi}{2}} = \frac{1}{2}\left(\ln\left(2\cdot\frac{\pi}{2}-\pi\right) - \ln(-\pi)\right)$$

$$= \frac{1}{2}\left(\ln 0 - \ln(-\pi)\right) = \frac{1}{2}\left(-\infty \ \dots\right) = -\infty.$$

c) $\displaystyle\int_1^\infty \frac{\ln x}{x\sqrt{x^2-1}}\,dx < \int_1^\infty \frac{1}{\sqrt{x^3-x}} < \infty$ divergent.

$$\int_1^\infty \frac{1}{x^p}\,dx \quad \begin{cases} \text{conv. if } p>1 \\ \text{div. if } p \le 1 \end{cases}$$

$\ln x < \sqrt{x} \ \Big| \cdot \dfrac{1}{x\sqrt{x^2-1}}$

$$\Rightarrow \frac{\ln x}{x\sqrt{x^2-1}} < \frac{\sqrt{x}}{x\sqrt{x^2-1}} = \frac{1}{\sqrt{x}\cdot\sqrt{x^2-1}} = \frac{1}{\sqrt{x^3-x}}$$

$\displaystyle\int_1^\infty \frac{1}{\sqrt{x^3-x}}\,dx$ „like" $\displaystyle\int_1^\infty \frac{1}{x^{\frac{3}{2}}}\,dx$ $\frac{3}{2}>1$ $\Rightarrow$ convergent.

⑤. Use the integral test to study the convergence.

a) $\displaystyle\sum_{n=2}^\infty \frac{1}{n(\ln n)^2}$ „like" $\displaystyle\int_2^\infty \frac{1}{x(\ln x)^2}\,dx.$

$\displaystyle\int_1^\infty f(x)\,dx$ „like" $\displaystyle\sum_{n=1}^\infty f(n)$

(if we only care about convergence, it does not matter where I start.)

$$\int_2^\infty \frac{1}{x(\ln x)^2}\,dx$$

$u = \ln x \Rightarrow du = \frac{1}{x}\,dx$

$x = 2 \Rightarrow u = \ln 2$

$x = \infty \Rightarrow u = \ln\infty = \infty$

$$\int_{\ln 2}^\infty \frac{1}{u^2}\,du = \int_{\ln 2}^\infty u^{-2}\,du = \frac{u^{-1}}{-1}\Big|_{\ln 2}^\infty = -\frac{1}{u}\Big|_{\ln 2}^\infty = -\frac{1}{\ln 2}$$ conv.

– We computed the actual value of the integral –

Ratio test $\to 1$

Cauchy test ... $\Big\}$

b) $\displaystyle\sum_{n=2}^{\infty} \frac{\ln n}{n^2}$ „like" $\displaystyle\int_2^{\infty} \frac{\ln x}{x^2}\,dx$

the problem

$$\int_2^{\infty} \frac{\ln x}{x^2}\,dx = \int_2^{\infty} \ln x \cdot \left(-\frac{1}{x}\right)' dx = -\frac{1}{x}\ln x \Big|_2^{\infty} + \int_2^{\infty} \frac{1}{x}\cdot\frac{1}{x}\,dx =$$

$$= \underbrace{-\lim_{x\to\infty} \frac{\ln x}{x}}_{0} + \frac{\ln 2}{2} + \int_2^{\infty} \frac{1}{x^2}\,dx = \frac{\ln 2}{2} + \int_2^{\infty} x^{-2}\,dx =$$

$$= \frac{\ln 2}{2} + \frac{x^{-1}}{-1}\Big|_2^{\infty} = \frac{\ln 2}{2} - \frac{1}{x}\Big|_2^{\infty} = \frac{\ln 2}{2} - \frac{1}{2} = \frac{\ln 2 - 1}{2}$$

c) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n\cdot \ln n \cdot \ln(\ln n)}$ „like" $\displaystyle\int_2^{\infty} \frac{1}{x \ln x \cdot \ln(\ln x)}\,dx$

$$(\ln(\ln x))' = \frac{1}{\ln x}\cdot\frac{1}{x} = \frac{1}{x\cdot \ln x}$$

Let $u = \ln(\ln(x))$

$\Rightarrow du = \dfrac{1}{x \ln x}\,dx$

$x = 2 \Rightarrow u = \ln(\ln 2)$

$x = \infty \Rightarrow u = \infty.$

$$\int_{\ln(\ln 2)}^{\infty} \frac{1}{u}\,du = \ln u \Big|_{\ln(\ln 2)}^{\infty} = \infty - \ln(\ln(\ln 2)) = \infty \text{ divergent.}$$